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HEATCAST NCR

Can machine learning predict tomorrow's heat level in Metro Manila better than copying today's?

CCMACLRL Introduction to Machine Learning · COM241ML · Dr. Armida P. Salazar

THE PROBLEM AND THE DATA

Next-day heat risk for the 16 cities of Metro Manila

Schools, employers of outdoor workers and city offices must decide a day ahead whether to move activities, prepare water and cooling, or warn residents.

We predict the PAGASA level of tomorrow's highest heat index for each city, from today's weather in that city.

The five PAGASA heat index levels

Not Hazardous	Caution	Extreme Caution	Danger	Extreme Danger
below 27 °C	27–32 °C	33–41 °C	42–51 °C	52 °C and up

16

cities of the National Capital Region, each with its own row

29,168

city-days from 2021 to 2025 (ERA5 weather via Open-Meteo)

5 + city

inputs about today: heat index, lowest temperature, rain, pressure, sunshine

1 day

ahead: the target is tomorrow's PAGASA level in that city

WHAT THE DATA SHOWED

Three findings, three decisions (training days only)

7.9%

of rows are Danger; 78.5% are Extreme Caution

So we judge models by macro F1 and Danger recall, not accuracy, and try class weights.

Apr–Jun

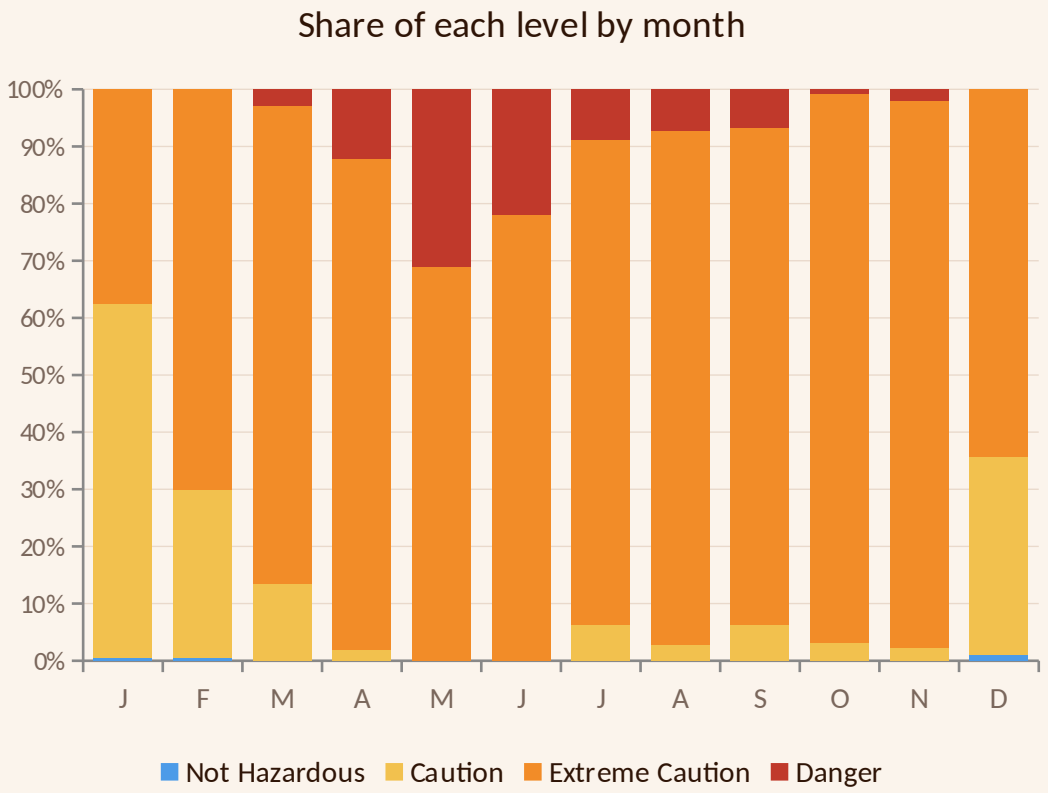
Danger peaks in April to June (31% of May rows) and never occurs in December to February

So the split follows time, never a random shuffle.

17% vs 2%

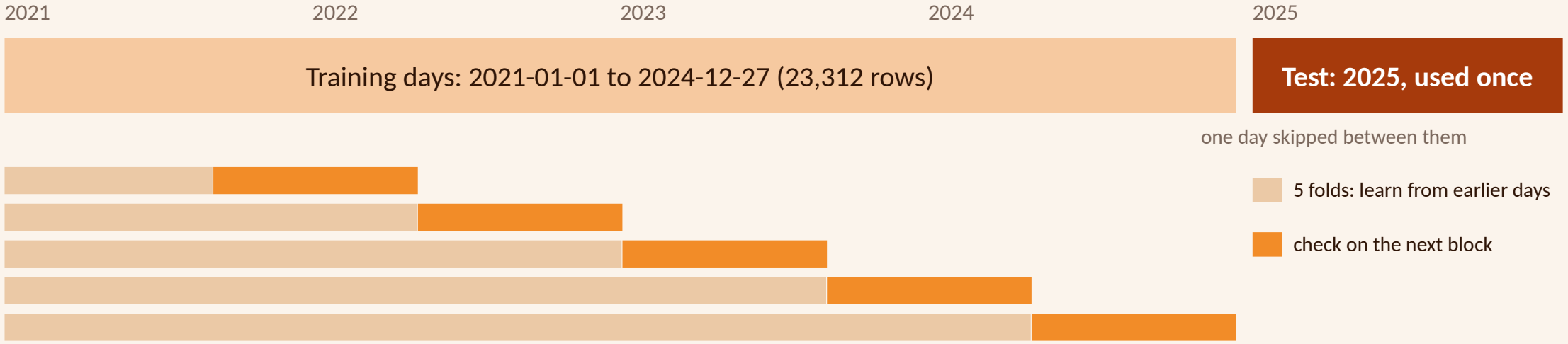
Danger rows in Malabon, Navotas and Valenzuela vs Muntinlupa and San Juan

So every city keeps its own row, and the city is an input.



HOW WE COMPARED FAIRLY

Same split, same folds and same scoring for every model



Primary metric

Macro F1 over Caution, Extreme Caution and Danger, so every level counts equally.

Selection rule

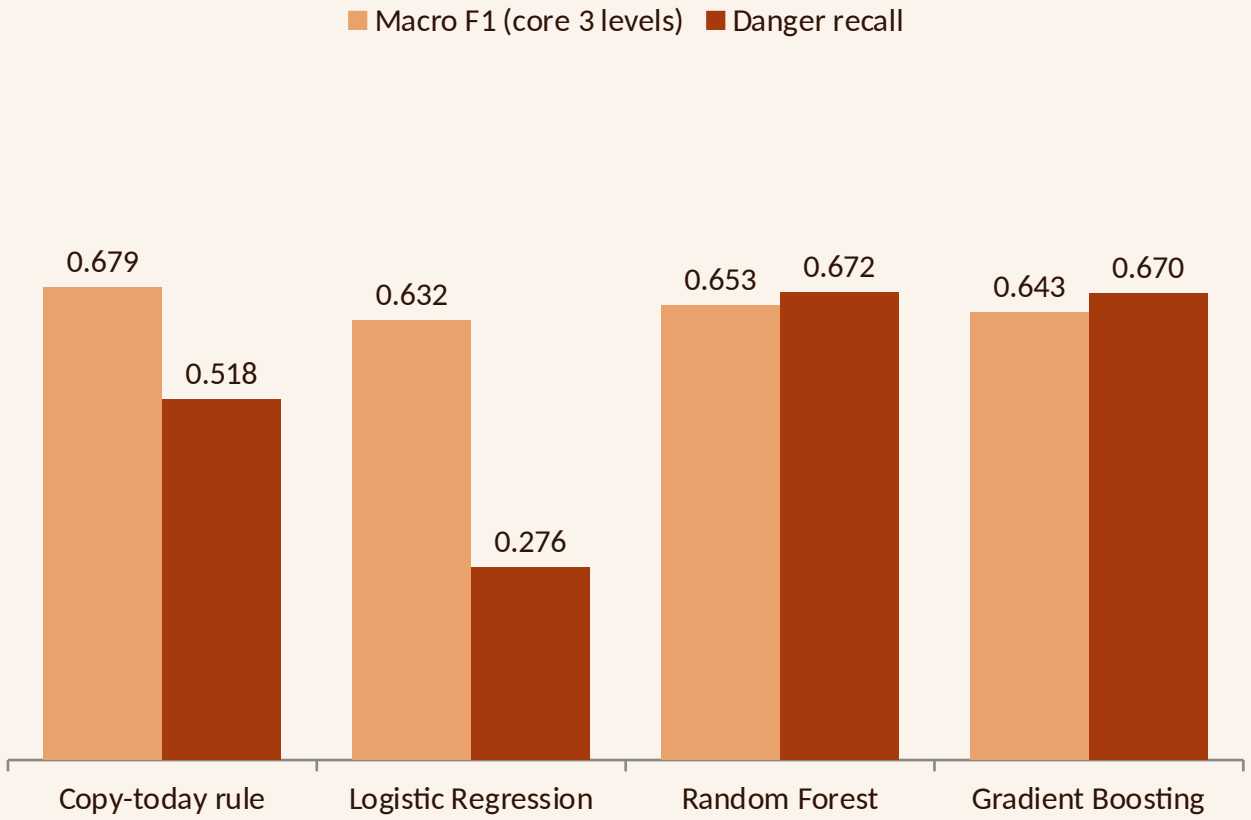
First catch more Danger days than the baseline; then the highest macro F1 wins.

Baseline to beat

Copy today: tomorrow's level is today's level in the same city.

THREE MODELS VS. THE RULE

Five-fold cross-validation on 2021 to 2024: mean over the folds



Logistic Regression Harry

C = 10, no class weights

Most precise, but caught only 28% of Danger days.

Random Forest Edrian

SELECTED

500 trees, depth 12, balanced weights

Best macro F1 of the three and 67% of Danger days caught.

Gradient Boosting Marc

200 trees, depth 2, balanced weights

Almost the same as Random Forest (0.643 vs 0.653).

FINAL TEST ON 2025

The selected Random Forest, run once on 365 untouched days

84%

of Danger days caught

293 of 347 · copy-today rule: 60%

36%

of Danger warnings right

293 of 816 · about 2 false alarms per hit

0.614

macro F1 (core 3 levels)

copy-today rule: 0.675 · accuracy 0.731 vs 0.824

An early-warning aid, not a better all-round forecaster: it catches far more Danger days than the rule, at the price of more false alarms.

LIVE DEMO

heatcast-ncr.streamlit.app

1

A hot day in Manila

Heat index 45.5 °C, lowest 29 °C, no rain, 1,008 hPa, 7,000 Wh/m²: Danger, about 90%

2

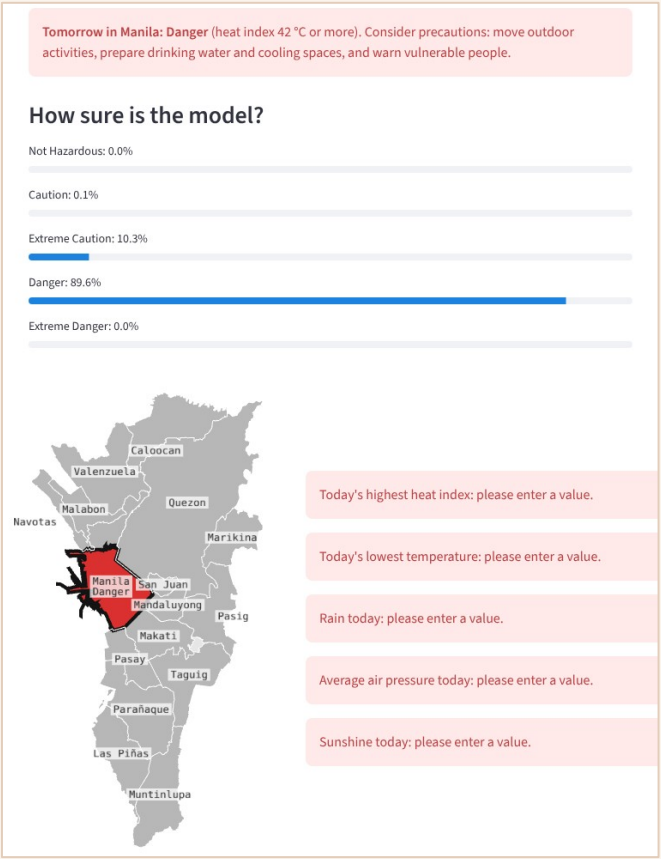
An impossible value

Rain of −5 mm: an error with the accepted range, and no prediction

3

An empty form

Every missing value is named; nothing is filled in silently



The app loads the saved Random Forest (the model we tested) and checks every input; 51 automated tests cover valid, invalid and boundary values.

CONCLUSION

What we learned

01

The answer

No model beat copying today's level on macro F1 or accuracy: tomorrow's heat is mostly today's heat.

02

What ML adds

With balanced class weights, the Random Forest caught 84% of the 2025 Danger days, so we deployed it as an early-warning aid.

03

Limits and next steps

Weather-model estimates, only five years. Next: month and recent-day inputs, a weather forecast input, more years, and a user-set Danger threshold.

THANK YOU

Questions?

heatcast-ncr.streamlit.app · github.com/keyyum/heatcast-ncr-simple